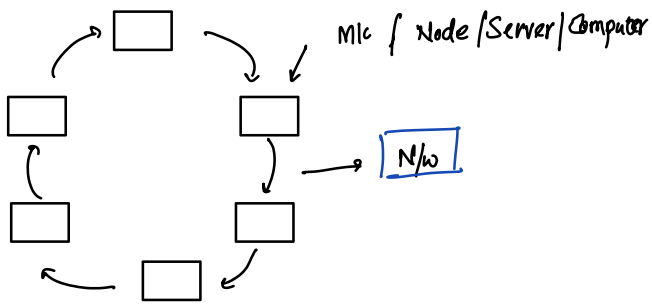
Distributed Systems → Group of computers.

Why we need it?

Data → Storage, Computation → can be done inside system - if small

But ↑↑↑ data → we need many systems connected over a n/w → DS.
(as we can't just one laptop anywhere)

(Data-driven decisions)
[Earlier → Based on intution of CEO etc)
↓
but now understood that do by customer's choices. → Data collection → used for taking business decisions.



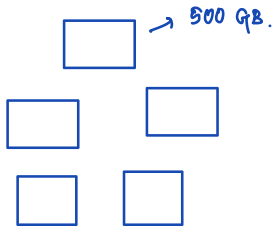
⇓

Objective: User should feel that they're using one system

Example: Distributed Storage

Need (2000 GB)

Thus, can do distributed storage

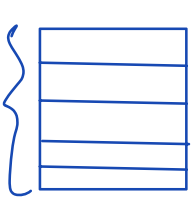


Distributed computation Ex: E-commerce: every 2 weeks settles payments with all merchants

Time Taking ← of millions of merchants for Amazon details.

↓

Divide into chunks and give to diff^t systems.



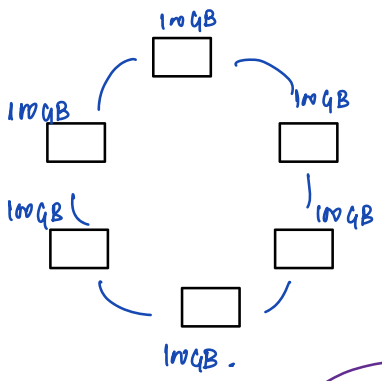
Principles

① Commodity Hardware → no single m/c becomes very expensive
↳ cheap/ Interchangeable

② N/w plays a very important role.

Evolution of Distributed systems - I (~ 2003) - IBM etc

data = 600 GB.
500 GB.
M/c



→ But, they failed miserably
↓
started losing data
↓
↓ Confidence

But, Google realized that going back is not an option.

Commodity hardware - so high chances of failing (but Oracle - expensive m/cs)

N/w - most unreliable thing - anything can go wrong

Google realized that few of the m/cs got disconnected due to some n/w issue - was not able to take data, thus, led to loss of data

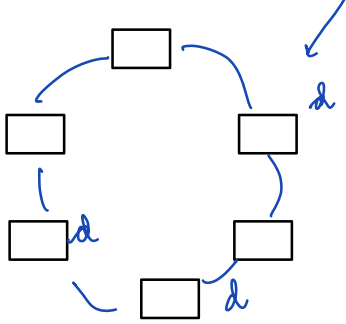
Evolution of Distributed Systems - II

System → Tolerant to these issues $\xrightarrow{\text{Solution}}$

① Redundancy / Replication

save multiple copies of data

→ solved problem of losing data.



storage (GFS) → Google File System (Distributed storage)

computation → Distributed Computation (Map Reduce) → Bigger problem to smaller problems so that they can be executed on multiple m/cs.

Became source of inspiration for creation of **Hadoop**

Centralized vs Decentralized Application (ownership-based)

① Centralized Application →

→ entity / group of entities (few) – control whole application – flow of data / info from single source

- single m/c / server
- Cluster of m/cs / servers.

Ex :→
- Facebook (own fleet of m/cs)
- Twitter
- Gmail

] 1 owner.

② Decentralized Application

→ No single authority

→ Distributed control

eg. Block chain , Bit torrent

consumer dependent

Where monolithic applications fit in?

can operate in both centralized / decentralized

(Fb/ LinkedIn / Paypal) started here this

eg. Bit Torrent.